🛒 Zepto SQL Data Analysis Project

📌 Project Overview

This project is a **SQL-based data analysis project** using a Zepto product dataset. The objective is to explore, clean, and analyze product data using **PostgreSQL** and generate meaningful business insights.

The project focuses on pricing, discounts, inventory, product categories, estimated revenue, and product value.

🗃️ Dataset

The project uses the `zepto_v2.csv` dataset containing product-level information from Zepto.

Main Columns

Column	Description
`sku_id`	Unique identifier for each product/SKU
`category`	Product category
`name`	Product name
`mrp`	Maximum Retail Price
`discountPercent`	Discount percentage
`availableQuantity`	Available inventory quantity
`discountedSellingPrice`	Selling price after discount
`weightInGms`	Product weight in grams
`outOfStock`	Indicates whether the product is out of stock
`quantity`	Product quantity

🛠️ Technologies Used

- **PostgreSQL**
- **SQL**
- **CSV**
- **GitHub**

🔍 Project Workflow

1. Data Exploration

The project begins by exploring the dataset and understanding its structure.

The analysis includes:

- Counting total records
- Viewing sample records
- Checking for NULL values
- Finding unique product categories
- Comparing in-stock and out-of-stock products
- Identifying duplicate products

2. Data Cleaning

The dataset is cleaned before performing business analysis.

The cleaning process includes:

- Identifying products with zero prices
- Removing invalid records where MRP is zero
- Converting price values from paise to rupees

Example:

```sql
UPDATE zepto
SET mrp = mrp / 100.0,
    discountedSellingPrice = discountedSellingPrice / 100.0;
```

📊 Business Insights

The project contains several business-oriented SQL queries.

🔹 Top 10 Best-Value Products

Identifies products offering the highest discount percentages.

🔹 High-MRP Products That Are Out of Stock

Finds expensive products that are currently unavailable.

🔹 Estimated Revenue by Category

Calculates estimated revenue based on selling price and available quantity.

🔹 Products With High MRP and Low Discount

Identifies products where:

- MRP is greater than ₹500
- Discount is less than 10%

🔹 Categories With the Highest Average Discount

Finds the top categories offering the highest average discount percentage.

🔹 Price Per Gram

Calculates the price per gram for products weighing at least 100 grams to identify better-value products.

🔹 Product Weight Classification

Products are classified into:

- **Low** — Less than 1000g
- **Medium** — 1000g to 4999g
- **Bulk** — 5000g or more

This is implemented using the SQL `CASE` statement.

🔹 Total Inventory Weight by Category

Calculates the total inventory weight available for each product category.

💡 SQL Concepts Demonstrated

This project demonstrates practical usage of:

```
CREATE TABLE
DROP TABLE
INSERT
UPDATE
DELETE
SELECT
WHERE
DISTINCT
```

```
ORDER BY
GROUP BY
HAVING
COUNT()
SUM()
AVG()
ROUND()
CASE
LIMIT
IS NULL
BOOLEAN filtering
```

📈 Skills Demonstrated

Through this project, I practiced:

- Writing SQL queries for data analysis
- Cleaning and preparing raw data
- Extracting business insights from datasets
- Using aggregate functions
- Working with categorical and numerical data
- Inventory analysis
- Revenue calculations
- Conditional logic using `CASE`
- Structuring a complete SQL analytics project

📂 Project Structure

```
SQL-QUERY-PROJECT/
│
├── SQL_QUERY_PROJECT.sql
├── zepto_v2.csv
└── README.md
```

🚀 How to Run the Project

Step 1: Install PostgreSQL

Install PostgreSQL and make sure it is available on your system.

Step 2: Create a Database

Create a new PostgreSQL database using **pgAdmin** or the PostgreSQL terminal.

Step 3: Add the Dataset

Place the following file in your project folder:

```
zepto_v2.csv
```

Step 4: Open the SQL File

Open:

```
SQL_QUERY_PROJECT.sql
```

Step 5: Run the SQL Script

Execute the script in **pgAdmin Query Tool** or another PostgreSQL-compatible SQL editor.

The SQL script performs:

1. Table creation
2. Data exploration
3. Data cleaning
4. Product analysis
5. Business insight generation

🎯 Project Objective

The main objective of this project is to demonstrate how **SQL can be used to transform raw e-commerce product data into meaningful business insights**.

The analysis focuses on:

- Product pricing
- Discounts
- Inventory
- Product categories
- Estimated revenue
- Product weight
- Product value

👨‍💻 Author

Anuraj Singh

Aspiring Full Stack Developer | SQL & Data Analysis Enthusiast

📧 **Email:** Anurajsingh597@gmail.com

🔗 **LinkedIn:** [Anuraj Singh]

⭐ Support

If you found this project useful or interesting, consider giving the repository a ⭐ on GitHub!